

NICOLE BISHOP-ADIGWE

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EDUCATION & CERTIFICATION

MEMORIAL UNIVERSITY

B.Eng. (Co-op) Computer Engineering

Sept 2020 - May 2026

AWS Web Services

AWS Certified Solutions Architect – Associate

In Progress

WORK EXPERIENCE

Software Developer Intern, Angler Solutions

Sept 2025 - Dec 2025

- Developed modular frontend with Vue.js/Nuxt.js using Apache ECharts for interactive graphs and heat maps; built reusable components with explicit loading, empty, and error states across energy modelling workflows.
- Built RESTful APIs with FastAPI, implemented API key-based authentication; used asyncio to cut response times by 75% under high-concurrency loads.
- Implemented a Python-based pumped-storage hydroelectric simulation model with automated execution and structured JSON export; validated outputs with pytest to prevent regressions.
- Designed scalable PostgreSQL database using SQLAlchemy to store scenario data and simulation results, improving query performance.

Software Developer Intern, Kraken Robotics

Jan 2025 - Apr 2025

- Developed reusable front-end components using React, TypeScript and styled components, contributing to the delivery of scalable internal software tools.
- Built RESTful APIs with Express.js and Node.js to automate backend script execution, eliminating manual data pipeline runs for end users.
- Integrated gRPC services using Protocol Buffers for low-latency communication between distributed frontend and backend services, supporting real-time battery diagnostics.
- Queried real-time battery hardware data using GraphQL and InfluxDB to support live diagnostic visualizations for battery monitoring system.
- Automated build tasks with Bash and updated GitLab pipelines to support CI/CD processes, using Git and SVN for version control.

Software Development Intern, Genoa Design International Ltd

Apr 2024 - Aug 2024

- Designed a PCN (Preliminary Change Notification) module within the web application, using React with TypeScript to improve UI responsiveness and workflow efficiency.
- Created RESTful APIs in C# with ASP.NET Core and used Swagger for testing PCN endpoints, enabling efficient front-end and back-end communication.
- Implemented automated API testing using C# and xUnit, to improve system stability and reduce production risks.
- Developed a SQL-based database in SQL Server, enabling efficient data storage, retrieval, and handling of PCN-related information.
- Collaborated on code reviews, branching strategies, and task tracking in Azure DevOps to support agile delivery.

Software Development Intern, Genoa Design International Ltd

Sept 2023 - Dec 2023

- Built and optimized AutoCAD and Navisworks 3D visualization plugins in C#, .NET, and WPF, developing XAML-based UI panels and dialogs that reduced load time for complex engineering models from over an hour to under five minutes.
- Programmed and tested a Navisworks viewpoint management system, diagnosing and resolving data processing issues and formally documenting defect reproduction and resolution.

PROJECTS

LLM-Assisted Electronic Design Automation Project

May 2025 - Apr 2026

- Designed an AI-powered EDA system using Python, PyTorch, and Llama-3 to generate SPICE-ready circuit parameters for analog amplifier design, reducing manual iteration from days to under an hour.
- Built a closed-loop pipeline integrating LLM outputs with Cadence simulations and output parsing to iteratively refine circuit performance across automated design cycles.

Job Application Project

Sept 2024 - Dec 2024

- Containerized a microservices application using Docker; built REST APIs with Express.js and added unit and integration tests, reinforcing configuration management and deployment reliability.

LEADERSHIP & TEAMS

Director of Events, NSBE MUN Chapter

Dec 2022 - Apr 2026

- Planned and executed technical networking events for industry partners (Ciena, Nokia e.t.c), leading logistics and student engagement over 3+ academic years

Paradigm Hyperloop - Electrical Team

July 2022 - Sept 2023

- Designed PCBs in KiCad, tested them, and programmed sensors to support autonomous vehicle navigation and control.
- Implemented SPI communication between sensors and the onboard microcontroller, supporting real-time data acquisition for autonomous navigation.

SKILLS

Languages & Scripting: Python, C#, Bash, PowerShell, C++, Java, JavaScript, TypeScript, SQL, XAML, HTML/CSS, VHDL, MATLAB

Frameworks & Tools: .NET, ASP.NET Core, WPF, React, Vue.js/Nuxt.js, Node.js, Express.js, FastAPI, AutoCAD, Navisworks, Git, SVN, Azure DevOps

Systems & Environments: Linux (Ubuntu), Windows Server/Desktop, Virtual Machines, Docker, AWS (EC2, S3, Lambda)

Databases: PostgreSQL, SQL Server, InfluxDB, SQLAlchemy

Testing & Automation: pytest, xUnit, Swagger, GitLab CI/CD | **AI/ML:** PyTorch, Llama-3, Prompt engineering, LLM APIs, Cadence simulations

Networking & Protocols: REST APIs, gRPC/Protocol Buffers, GraphQL, TCP/IP | **Other:** Agile/Scrum, MS Office Suite (Excel, Word, PowerPoint)